

INVENTORY AND WAREHOUSE MANAGEMENT SYSTEM

TEST PLAN AND TEST CASES

Course:

SEN201 – Software Engineering

Department:

Department of Computer Science

Academic Session:

2025/2026

1. INTRODUCTION

This document describes the testing procedures and test cases used to evaluate the Inventory and Warehouse Management System. The purpose of testing is to verify that the system functions correctly and satisfies the specified requirements.

The testing process focuses on validating inventory operations, warehouse activities, authentication, reporting, and stock monitoring functionalities.

2. TESTING OBJECTIVES

The objectives of testing are:

1. To verify that all system modules function correctly.
 2. To identify and reduce system errors.
 3. To validate inventory processing operations.
 4. To ensure accurate stock monitoring and reporting.
 5. To confirm that role-based authentication works properly.
 6. To improve system reliability and usability.
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3. TESTING SCOPE

The following modules were tested:

- Authentication Module
 - Product and Category Management
 - Goods Receipt Processing
 - Goods Dispatch Processing
 - Inventory Monitoring
 - Low Stock Alert System
 - Reports Module
 - Staff Management
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4. TESTING STRATEGY

The system was tested using functional testing techniques. Each module was tested individually to ensure that expected outputs matched actual system behavior.

The testing process involved:

- Valid input testing
 - Invalid input testing
 - Inventory transaction testing
 - Authentication testing
 - Reporting and dashboard testing
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5. TEST CASES

Test ID	Module	Test Action	Expected Result	Actual Result	Status
TC-001	Login	Enter valid email and password	User logs into dashboard successfully	User logged in successfully	Pass
TC-002	Login	Enter invalid password	Error message displayed	Error message displayed	Pass
TC-003	Product Management	Add new product with SKU	Product saved successfully	Product saved successfully	Pass
TC-004	Category Management	Add new category	Category created successfully	Category created successfully	Pass

Test ID	Module	Test Action	Expected Result	Actual Result	Status
TC-005	Goods Receipt	Record received stock	Stock quantity increases	Stock updated correctly	Pass
TC-006	Goods Dispatch	Dispatch available stock	Stock quantity reduces	Stock reduced correctly	Pass
TC-007	Inventory Monitoring	View inventory dashboard	Inventory data displayed	Inventory data displayed correctly	Pass
TC-008	Low Stock Alert	Reduce stock below threshold	Low-stock alert displayed	Alert displayed successfully	Pass
TC-009	Reports	Generate inventory report	Report generated successfully	Report generated correctly	Pass
TC-010	Staff Management	Create staff account	Staff account created successfully	Account created successfully	Pass

6. TEST SUMMARY

The Inventory and Warehouse Management System was tested successfully across all major modules. The system performed inventory operations correctly and generated accurate stock updates, reports, and low-stock alerts.

Authentication, goods receipt, goods dispatch, reporting, and inventory monitoring functionalities operated according to the expected requirements.

The testing process confirmed that the system is functional, reliable, and suitable for warehouse inventory management operations.

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RISK REGISTER

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1. INTRODUCTION

This document identifies potential risks associated with the development and implementation of the Inventory and Warehouse Management System.

Risk Exposure (RE) is calculated using the formula:

$$RE = \text{Probability} \times \text{Consequence}$$

Where:

- Probability represents the likelihood of the risk occurring.
 - Consequence represents the impact of the risk on the project.
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2. RISK REGISTER TABLE

Risk ID	Risk Description	Probability	Consequence	RE	Mitigation Strategy
R-001	Database failure	0.4	8	3.2	Perform regular database backups
R-002	Team member unavailable	0.3	6	1.8	Share project tasks among members
R-003	Poor internet connection	0.5	5	2.5	Use local development environment
R-004	Incorrect stock entry	0.4	7	2.8	Validate inventory input data
R-005	Unauthorized system access	0.2	9	1.8	Implement authentication and RBAC
R-006	Late project completion	0.3	8	2.4	Follow project timeline strictly

Risk ID	Risk Description	Probability	Consequence	RE	Mitigation Strategy
R-007	Loss of project files	0.2	8	1.6	Use Git and cloud backup storage
R-008	System bugs during demo	0.3	7	2.1	Conduct proper testing before presentation

3. RISK SUMMARY

The major risks identified in the project include database failure, incorrect stock entries, internet connectivity issues, and implementation delays.

The project team minimized these risks through proper testing, version control, authentication mechanisms, data validation, and regular project backups.

The implementation of mitigation strategies reduced the overall impact of potential risks on the Inventory and Warehouse Management System project.